Introduction to Logic (CS) 2026-2027
Homework Assignment 2

Submission deadline: Wednesday, 21st October, 09:00 AM

This homework assignment consists of 6 questions, plus a bonus question. Each question indicates how many points it is worth. With the regular questions, you can earn 100 points. With the bonus question, you can earn additionally 10 points. The grade is the number of points you earned divided by 10, with a maximum of 10.

The submissions instructions are in BrightSpace (Content → Homework Assignments → General Information), **please read them carefully and in full**.

This is an individual assignment. Do not copy your homework from someone (or something) else (this includes LLMs), and do not share your solutions with anyone! If fraud is suspected (e.g., if your answers are too similar compared to the answers of someone else) the Board of Examiners will be notified.

Questions

1. (22 points) Translate the following sentences into *first-order logic*. The domain of discourse is the set of all students. There is one translation key for the entire question which contains the unary function

$$s(x): \text{ the sister of } x$$

   (a) Provide the full translation key for the sentences below.

   (b) Translate the following sentence into FOL, using the translation key you provided above: *Elitsa is the only student who helps their sister.*

   (c) Translate the following sentence into FOL, using the translation key you provided above: *Anyone who helps Kester is being helped by someone who plays chess.*

   (d) Translate the following sentence into FOL, using the translation key you provided above: *Nobody helps anyone who doesn't help themselves.*

2. (30 points) Give formal proofs, including justifications, for the following inferences. You can only use the Reiteration rule and the Introduction and Elimination rules from system $\mathcal{F}$.

   (a) (8 points)

$$\vdash \neg\exists y\forall x(\neg K(x,x) \leftrightarrow K(y,x))$$

   (b) (10 points)

$$\begin{array}{l} \forall x\forall y(R(x,y) \rightarrow R(y,x)) \\ \forall x\forall y\forall z((R(x,y) \wedge R(y,z)) \rightarrow R(x,z)) \\ \hline \forall x(\exists y R(x,y) \rightarrow R(x,x)) \end{array}$$

(c) (12 points)

$$\begin{array}{|l} \exists x(P(x) \wedge (Q(x) \vee (x = m(x)))) \\ \forall x(Q(x) \rightarrow P(m(x))) \\ \hline \exists x(P(x) \wedge P(m(x))) \end{array}$$

3. (8 points) Let the following sets be given:

$$A = \{\emptyset, \{1,2\}, \{2,1\}, 1\}, \quad B = \{1, \{\emptyset\}, \{1,2\}\}, \quad C = \{\emptyset, R\}, \quad R = \{\langle 0,1\rangle, \langle 1,2\rangle, \langle 2,0\rangle\}$$

For each of the following statements, determine whether it is true or false. Briefly justify your answers.

(a) A has four elements ($|A| = 4$)
(b) $\{1\} \in A$
(c) $\{\emptyset\} \in B$
(d) $A = B$
(e) $R \in C$
(f) $\langle 1,2\rangle \in C$
(g) R represents a total unary function on the set $\{0,1,2\}$
(h) R is a transitive relation on $\{0,1,2\}$

4. (15 points) Let $\mathcal{L}$ be the first-order language with one constant a, one unary function s, one unary predicate E, one binary predicate L.

Let $\mathfrak{M}$ be the $\mathcal{L}$-model with:

- domain $\mathfrak{M}(\forall) = D = \{0, 1, 2, \ldots, 11\}$,
- $\mathfrak{M}(a) = 0$,
- $\mathfrak{M}(s)(x) = x + 1 \mod 12$, e.g. $\mathfrak{M}(s)(0) = 1$, $\mathfrak{M}(s)(1) = 2$, ..., $\mathfrak{M}(s)(10) = 11$, $\mathfrak{M}(s)(11) = 0$,
- $\mathfrak{M}(E) = \{0, 2, 4, 6, 8, 10\}$,
- $\mathfrak{M}(L) = \{\, \langle x, y\rangle \in D \times D \mid x < y \,\}$

Let h be a variable assignment with $h(x) = 11$, $h(y) = 4$. For each of the following statements, determine whether it is true or false. Motivate your answers by following the definition of satisfaction step by step. Make sure to explain your reasoning. This can be done by referring to the satisfaction clause that you used in each step (e.g. giving justification (sat. $\rightarrow$)), or by providing explanatory text that forms a coherent narrative/argumentation, e.g., as in the short solution to part (c) in `semanticsWorkout.pdf`.

(a) $\mathfrak{M} \models \big(\forall x \exists y(s(y) = x \wedge (L(y,x) \vee E(x)))\big)[h]$
(b) $\mathfrak{M} \models \big(\forall y(\neg L(x,y) \rightarrow \exists z(s(x) = z \wedge L(y,z)))\big)[h]$

5. (10 points) Consider the following situation:

Several students are cycling to the Noorderplantsoen. Andy and Sam have a flat tyre. Charles is carrying a cello and a bottle of wine. Eloise is carrying a bottle of wine. She and David are first-year students. Charles is older than Andy, and both Sam and Andy are older than Eloise. David is the youngest of them all.

Let $\mathcal{L}$ be the first-order language arising from the following vocabulary:

unary predicate $F(x)$:	x has a flat tyre
unary predicate $C(x)$:	x is carrying a cello
unary predicate $W(x)$:	x is carrying a bottle of wine
unary predicate $S(x)$:	x is a first-year student
binary predicate $O(x,y)$:	x is older than y

(a) Define a first-order $\mathcal{L}$-model $\mathfrak{M}$ that describes the situation described above, where the elements of $\mathfrak{M}(\forall)$ are students. In the model, the interpretation of the relation O should be transitive.

(b) Give sentences A and B in the first-order language $\mathcal{L}$ such that each sentence contains at least two nested quantifiers (i.e. one quantifier is in the scope of another), and moreover, A is true and B is false in your model. Note that $\mathcal{L}$ does not contain any constants, and you cannot change $\mathcal{L}$ by adding constants.

6. (15 points) Consider the following argument:

Every robot is charging at exactly one station.
At every station, at least two robots are charging.
No robot is a station.
∴ Any two different robots are charging at the same station.

(a) Define a first-order vocabulary (i.e., a translation key) for expressing the argument in first-order logic. State explicitly for each item whether it is a constant, function symbol or predicate and include its arity.

(b) Let us refer to the first-order language arising from the vocabulary you specified in part (a) as $\mathcal{L}$. Translate the argument into your first-order language $\mathcal{L}$.

(c) Define a countermodel for the argument. That is, define a first-order $\mathcal{L}$-model $\mathfrak{M}$ such that all premises are satisfied in $\mathfrak{M}$, but the conclusion is not. The model should contain *at most* 8 elements in the domain. (You do not have to demonstrate or argue that your model is, in fact, a countermodel.)

7. (Bonus question, 10 points) Give a formal proof, including justifications, for the following inference. You can only use the Reiteration rule and the Introduction and Elimination rules from system $\mathcal{F}$.

$$\begin{array}{|l} \neg\forall x\, P(x) \\ \forall x(\neg P(x) \to \exists y\, Q(y)) \\ \hline \exists x\, Q(x) \end{array}$$